HOORVIA - PEMDAS ORDER OF OPERATIONS CHART: PRINTABLE MATH STEPS & EXAMPLES

Quick Reference Cheat Sheet · US Letter Printable

Reference Charts · Sheet #1

Step	Operation	Rule / Example	Worked Result
P - Parentheses	( ) [ ] { }	Solve inside grouping symbols first	(3 + 2) x 4 = 20
E - Exponents	^{2} ^{3} ?	Powers and roots, left to right	$2^3 = 8$
M - Multiplication	x	With division, left to right - not before it	6 x 2 = 12
D - Division	÷ /	With multiplication, left to right	12 ÷ 3 = 4
A - Addition	+	With subtraction, left to right - not before it	5 + 3 = 8
S - Subtraction	?	With addition, left to right	9 ? 4 = 5
Example 1	2 + 3 x 4	Multiply first: 3 x 4 = 12, then 2 + 12	Answer: 14 (not 20)
Example 2	(2 + 3) x 4	Parentheses first: 5 x 4	Answer: 20
Example 3	10 ? 2^2	Exponent first: 10 ? 4	Answer: 6
Example 4	8 ÷ 2 x 4	Left to right: 4 x 4	Answer: 16 (not 1)
Example 5	3 x 2^3	Exponent first: 3 x 8	Answer: 24
Example 6	(6 + 2) ÷ 2^2	Parentheses, then exponent: 8 ÷ 4	Answer: 2

PRO TIP: M and D are equals: Multiplication and division share one step - work them left to right, never M before D.